

# KHAIR UL WARA HUSSAIN

AI/ML Engineer | Python Developer

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## PROFILE

AI/ML and full-stack engineer with hands-on experience building NLP, RAG, and computer-vision systems, automated data pipelines, and production MLOps infrastructure. Skilled at translating research and messy data into deployable tools, dashboards, and applications, with a track record of taking projects end-to-end from research to working product.

## EDUCATION

BS in Artificial Intelligence — COMSATS University Islamabad

2023 – 2027 (Expected)

## WORK EXPERIENCE

AI Intern — AKSA SDS (Remote)

Jul 2026 – Aug 2026

- Fine-tuned a DeBERTa-v3-base transformer model for an enterprise adverse media screening system, achieving 95% accuracy on a held-out evaluation set through threshold calibration and aggregation-strategy validation.
- Designed and implemented a weak-supervision and LLM-assisted labeling pipeline using sanctions and PEP data sources, scaling a QA'd labeled training dataset to 2,200+ rows.
- Conducted regional performance analysis and augmented the training corpus with 700+ Pakistan-context articles to improve model accuracy on Pakistani media sources.
- Collaborated cross-functionally with technical stakeholders and delivered structured weekly R&D progress reports.

Python Web Scraping Developer — Tayton Capital, LLC (Remote)

Mar 2026 – May 2026

- Engineered Python/Playwright pipelines extracting 10,000+ real estate records daily, reducing per-record latency by ~35%.
- Implemented proxy rotation and anti-bot measures, achieving 99%+ pipeline uptime across dynamic data sources.
- Built ETL workflows delivering structured property data to CSV, cutting manual data prep time by 80%.

AI/ML Engineering Intern — DevelopersHub Corporation

Feb 2026 – Mar 2026

- Built a RAG-based chatbot (LangChain + Streamlit) delivering context-accurate responses over a 500+ document corpus.
- Fine-tuned BERT/BART for multi-class ticket categorization (91% accuracy across 10+ categories); built Scikit-learn pipelines with GridSearchCV for churn prediction, improving F1-score by 18% over baseline.

## PROJECTS

Self-Healing MLOps Pipeline for Sentiment Analysis | Jenkins, Docker, Kubernetes, Prometheus, Grafana, AWS EC2

- Built a self-healing CI/CD and MLOps pipeline that auto-rolls back a sentiment-analysis API to a stable model via blue-green Kubernetes deployments on confidence drift.

LeafLogic: AI Plant Disease Diagnostic | Flutter, CNN, TensorFlow

- Built a Flutter CNN app diagnosing 38 plant diseases from leaf images with 88% accuracy on the PlantVillage dataset.

Context-Aware AI Chatbot | LangChain, ChromaDB, Streamlit, RAG

- Built a LangChain RAG chatbot that retrieves and grounds answers in ingested documents while preserving multi-turn chat history.

News Classifier | BERT, PyTorch, Hugging Face Transformers

- Fine-tuned BERT on the AG News corpus for 4-way topic classification, achieving 89% accuracy and weighted F1.

Auto-Tagging Support Tickets Using LLM | LLM, Zero/Few-Shot Prompting, LoRA Fine-Tuning, Scikit-learn

- Built an LLM ticket-tagging system comparing zero-shot, few-shot, and LoRA fine-tuning against a supervised baseline for top-3 tag prediction.

## VOLUNTEER EXPERIENCE

Digital Media & Outreach Coordinator — East Berkshire PSC, London

Sep 2023 – Present

- Manage event postings on the organization's website, keeping members and the wider community informed of upcoming activities and programs.
- Design event posters, flyers, and promotional visuals used across campaigns and outreach initiatives.
- Produce and distribute a monthly newsletter, curating content that keeps members engaged with organizational news and events.

## TECHNICAL SKILLS

- AI / ML:** Scikit-learn, LangChain, BERT, BART, RAG, Fine-tuning, CNNs, Random Forest, Isolation Forest, Time-Series Forecasting, Ensemble Methods
- Data Science:** Pandas, NumPy, Matplotlib, EDA, Feature Engineering, Geospatial Analysis, Statistical Modeling
- Backend:** FastAPI, REST APIs, SQLAlchemy, SQLite, Uvicorn, Pydantic, ETL Pipelines
- Frontend:** React (Vite), TypeScript, HTML, CSS, Leaflet/OpenStreetMap, Responsive Design
- Languages:** Python, TypeScript, SQL, Dart/Flutter, C++, Java
- Scraping:** Playwright, Selenium, Proxy Rotation, Anti-Bot Measures
- DevOps & Cloud:** Docker, Kubernetes (Minikube, Deployments, Services, PVCs, Blue-Green), Jenkins CI/CD, Prometheus, Grafana, Alertmanager, AWS (EC2, S3), GitHub Webhooks, Linux
- Tools:** VS Code, Jupyter, Google Colab, Git, MySQL, Streamlit, Gradio

## CERTIFICATIONS

Cisco: Python Essentials 1 & 2 • Introduction to Modern AI • Junior Cybersecurity Analyst Career Path • HTML & CSS Essentials